

Implement any one of the following Expert System

- I. Information management
 - II. Hospitals and medical facilities
 - III. Help desks management
 - IV. Employee performance evaluation
 - V. Stock market trading
 - VI. Airline scheduling and cargo schedules
-

```
package AI;
```

```
import java.util.ArrayList;
```

```
import java.util.Scanner;
```

```
class Patient {
```

```
    String name;
```

```
    int age;
```

```
    String[] symptoms;
```

```
    String diagnosis;
```

```
    String doctor;
```

```
    String medicine;
```

```
    Patient(String name, int age, String[] symptoms) {
```

```
        this.name = name;
```

```
        this.age = age;
```

```
        this.symptoms = symptoms;
```

```
    }
```

```
    void displayInfo() {
```

```
        System.out.println("\nPatient Name: " + name);
```

```
        System.out.println("Age: " + age);
```

```
        System.out.print("Symptoms: ");
```

```
        for (String symptom : symptoms) {
```

```
            System.out.print(symptom + ", ");
```

```
        }
```

```
        System.out.println("\nDiagnosis: " + (diagnosis != null ? diagnosis : "Pending"));
```

```
        System.out.println("Assigned Doctor: " + (doctor != null ? doctor : "Not Assigned"));
```

```
        System.out.println("Suggested Medicine: " + (medicine != null ? medicine : "Not Prescribed"));
```

```
    }
```

```
}
```

```
class HospitalExpertSystem {
```

```
    ArrayList<Patient> patients = new ArrayList<>();
```

```
    void addPatient(Patient patient) {
```

```
        patients.add(patient);
```

```
        diagnosePatient(patient);
```

```
    }
```

```
    void diagnosePatient(Patient patient) {
```

```
        int feverCount = 0, coughCount = 0, fatigueCount = 0, headacheCount = 0, stomachPainCount = 0;
```

```
        // Count symptoms with flexibility
```

```
        for (String symptom : patient.symptoms) {
```

```
            String s = symptom.toLowerCase().trim(); // Case-insensitive and trim spaces
```

```
            if (s.contains("fever")) feverCount++;
```

```
            if (s.contains("cough") || s.contains("cough")) coughCount++; // Handle typo
```

```
            if (s.contains("fatigue")) fatigueCount++;
```

```
            if (s.contains("headache")) headacheCount++;
```

```

        if (s.contains("stomach") || s.contains("stomach pain")) stomachPainCount++;
    }

    // Debugging: Print symptom counts
    System.out.println("Debug - Symptom Counts: Fever=" + feverCount + ", Cough=" + coughCount +
        ", Fatigue=" + fatigueCount + ", Headache=" + headacheCount +
        ", Stomach Pain=" + stomachPainCount);

    // Rule-based diagnosis
    if (feverCount > 0 && coughCount > 0 && fatigueCount > 0) {
        patient.diagnosis = "Influenza (Flu)";
        patient.doctor = "Dr. Sharma (General Physician)";
        patient.medicine = "Paracetamol, Cough Syrup";
    } else if (feverCount > 0 && headacheCount > 0) {
        patient.diagnosis = "Migraine or Viral Fever";
        patient.doctor = "Dr. Gupta (Neurologist)";
        patient.medicine = "Ibuprofen";
    } else if (stomachPainCount > 0 && fatigueCount > 0) {
        patient.diagnosis = "Gastritis or Food Poisoning";
        patient.doctor = "Dr. Patel (Gastroenterologist)";
        patient.medicine = "Antacid, Hydration Fluids";
    } else if (coughCount > 0 && fatigueCount > 0) {
        patient.diagnosis = "Common Cold";
        patient.doctor = "Dr. Singh (General Physician)";
        patient.medicine = "Cough Drops, Rest";
    } else if (feverCount > 0 && stomachPainCount > 0) { // New rule for fever + stomach pain
        patient.diagnosis = "Possible Stomach Infection or Fever";
        patient.doctor = "Dr. Patel (Gastroenterologist)";
        patient.medicine = "Antacid, Paracetamol";
    } else if (coughCount > 0) { // Single symptom rule
        patient.diagnosis = "Mild Cough";
        patient.doctor = "Dr. Singh (General Physician)";
        patient.medicine = "Cough Drops";
    } else {
        patient.diagnosis = "Unable to determine. Consult a specialist.";
        patient.doctor = "Not Assigned";
        patient.medicine = "Not Prescribed";
    }
}

void displayAllPatients() {
    if (patients.isEmpty()) {
        System.out.println("No patients in the system.");
        return;
    }
    System.out.println("\n--- All Patients in Hospital System ---");
    for (Patient patient : patients) {
        patient.displayInfo();
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        HospitalExpertSystem hospital = new HospitalExpertSystem();

        System.out.println("Welcome to Advanced Hospital Expert System");
        System.out.println("-----");

        while (true) {
            System.out.println("\n1. Add New Patient");
            System.out.println("2. Display All Patients");

```

```

System.out.println("3. Exit");
System.out.print("Enter your choice: ");
int choice = scanner.nextInt();
scanner.nextLine(); // Clear buffer

switch (choice) {
    case 1:
        System.out.print("Enter patient name: ");
        String name = scanner.nextLine();

        System.out.print("Enter patient age: ");
        int age = scanner.nextInt();
        scanner.nextLine(); // Clear buffer

        System.out.print("Enter symptoms (comma-separated, e.g., fever, cough, fatigue, headache, stomach pain):");
        String symptomsInput = scanner.nextLine();
        String[] symptoms = symptomsInput.split(",\\s*");

        Patient patient = new Patient(name, age, symptoms);
        hospital.addPatient(patient);
        System.out.println("Patient added successfully!");
        patient.displayInfo();
        break;

    case 2:
        hospital.displayAllPatients();
        break;

    case 3:
        System.out.println("Exiting system. Goodbye!");
        scanner.close();
        return;

    default:
        System.out.println("Invalid choice. Try again.");
}
}
}
}

```

.....

output

.....

Welcome to Advanced Hospital Expert System

1. Add New Patient

2. Display All Patients

3. Exit

Enter your choice: 1

Enter patient name: arati

Enter patient age: 35

Enter symptoms (comma-separated, e.g., fever, cough, fatigue, headache, stomach pain): cough

Debug - Symptom Counts: Fever=0, Cough=1, Fatigue=0, Headache=0, Stomach Pain=0

Patient added successfully!

Patient Name: arati

Age: 35

Symptoms: cough,

Diagnosis: Mild Cough
Assigned Doctor: Dr. Singh (General Physician)
Suggested Medicine: Cough Drops

1. Add New Patient
2. Display All Patients
3. Exit

Enter your choice: 1

Enter patient name: priya

Enter patient age: 30

Enter symptoms (comma-separated, e.g., fever, cough, fatigue, headache, stomach pain):
fever, stomach pain

Debug - Symptom Counts: Fever=1, Cough=0, Fatigue=0, Headache=0, Stomach Pain=1

Patient added successfully!

Patient Name: priya

Age: 30

Symptoms: fever, stomach pain,

Diagnosis: Possible Stomach Infection or Fever

Assigned Doctor: Dr. Patel (Gastroenterologist)

Suggested Medicine: Antacid, Paracetamol

1. Add New Patient
2. Display All Patients
3. Exit

Enter your choice: 2

--- All Patients in Hospital System ---

Patient Name: arati

Age: 35

Symptoms: cough,

Diagnosis: Mild Cough

Assigned Doctor: Dr. Singh (General Physician)

Suggested Medicine: Cough Drops

Patient Name: priya

Age: 30

Symptoms: fever, stomach pain,

Diagnosis: Possible Stomach Infection or Fever

Assigned Doctor: Dr. Patel (Gastroenterologist)

Suggested Medicine: Antacid, Paracetamol